

Multiple Choice (10 marks)

1. Which of the following instruction-set features is NOT generally considered an obstacle to aggressive pipelining of an integer unit?
 - (a) Condition codes set by every instruction
 - (b) Variable-length encoding of instructions
 - (c) Instructions requiring widely varying numbers of cycles to execute
 - (d) Several different classes (sets) of registers
2. What's the maximum number of edges in a simple triangle free planar graph with 30 vertices?
 - (a) 69
 - (b) 60
 - (c) 63
 - (d) 84
 - (e) 45
3. Adding more basis functions in a linear model, pick the most probably option:
 - (a) Decreases model bias
 - (b) Increases estimation bias
 - (c) Doesn't affect bias and variance
 - (d) Increases both bias and variance
 - (e) Decreases variance
4. Determine the output of the following program
`READ (2, 3) I, J 3FORMAT (2I11) IF (I - J) 5,6,6 6WRITE (3,10) I 10FORMAT (1X, I11) GO TO 7 5WRITE (3,10) J 7STOP`
 - (a) Prints the absolute value of the difference of two numbers
 - (b) Prints out the result of multiplying the two numbers
 - (c) Prints the second number twice
 - (d) Calculates the difference of two numbers
 - (e) Calculates the sum of two numbers
5. When the data must arrive at the receiver exactly as they were sent, its called
 - (a) Message Security

- (b) Message Clarity
- (c) Message Integrity
- (d) Message Authenticity
- (e) Message Splashing

True / False (10 marks)

Answer True or False

- 6. True or False: A MAC can be secure as long as the key length is sufficiently long, regardless of the message length.
- 7. True or False: There are 520 labelled forests on 8 vertices with 5 connected components, with vertex labels 1 to 8.
- 8. True or False: Bayesians and frequentists agree on the use of maximum likelihood estimation for parameter estimation.
- 9. True or False: Bitmap graphics cannot support millions of colors due to their inherent limitations in color depth.
- 10. True or False: Hidden Markov models are primarily used to analyze and predict stock market trends.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Write a function to sort a tuple by its float element.
- 12. Write a function to replace blank spaces with any character in a string.

End of Paper